

Initialize Reactive Jupyter | Sync all State code

```
import numpy as np
from scipy.optimize import linprog
```

Python

1. (10 points) Solve the following problems by the dual simplex method.

$$\begin{aligned} \max \quad & -8x_1 - 2x_2 - 12x_3 \\ \text{s.t.} \quad & -2x_1 - x_2 - 4x_3 \leq 1 \\ & -x_1 + 2x_2 - 2x_3 \leq -6 \\ & -x_2 + x_3 \leq 2 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

```
c = -1 * np.array([-8, -2, -12])
A = np.array([[ -2, -1, -4],
              [ -1, 2, -2],
              [ 0, -1, 1]])
b = np.array([1, -6, 2])
x_bounds = np.array([(0, None), (0, None), (0, None)])

# dual simplex
solution = linprog(c, A_ub=A, b_ub=b, bounds=x_bounds, method='highs-ds')

print(solution.message)
print(f'x1= {solution.x[0]}')
print(f'x2= {solution.x[1]}')
print(f'x3= {solution.x[2]}')
print(f'max Z: {-solution.fun}')
```

Python

```
... Optimization terminated successfully. (HiGHS Status 7: Optimal)
x1= 2.0
x2= 0.0
x3= 2.0
max Z: -40.0
```

2. (10 points) Use the dual simplex method to show that the following problem is infeasible.

$$\begin{aligned} \max \quad & -8x_1 - x_2 - x_3 \\ \text{s.t.} \quad & -2x_1 - x_2 \leq -20 \\ & 2x_1 - x_2 + x_3 \leq -12 \\ & x_1 + x_3 \leq 7 \\ & -x_1 + x_2 - x_3 \leq -5 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

```
c = -1 * np.array([-8, -1, -1])
A = np.array([[ -2, -1, 0],
              [ 2, -1, 1],
              [ 1, 0, 1],
              [ -1, 1, -1]])
b = np.array([-20, -12, 7, -5])
x_bounds = np.array([(0, None), (0, None), (0, None)])

sol = linprog(c, A_ub=A, b_ub=b, bounds=x_bounds, method="highs-ds")

print(sol.message)
```

Python

```
... The problem is infeasible. (HiGHS Status 8: model_status is Infeasible; primal_status is At lower/fixed bound)
```

$$\begin{aligned}
 \max \quad & z = 10x_1 + 7x_2 + 6x_3 && \text{(Profit \$)} \\
 \text{s.t.} \quad & 3x_1 + 2x_2 + x_3 \leq 36 && \text{(Resource 1)} \\
 & x_1 + x_2 + 2x_3 \leq 32 && \text{(Resource 2)} \\
 & 2x_1 + x_2 + x_3 \leq 22 && \text{(Resource 3)} \\
 & x_1, x_2, x_3 \geq 0
 \end{aligned}$$

	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	3	0	0	1	0	5	146
x_2	0	1	1	0	1	0	-1	14
x_5	0	-2	0	0	1	1	-3	2
x_3	0	1	0	1	-1	0	2	8

The following three questions are independent of each other.

- (a) (10 points) Suppose that the profit of producing one unit of Product 3 goes down from \$6 to \$3. Use sensitivity analysis to find a new production plan for the company.

Change in objective coeff: $3 - 6 = -3$

Objective Row

Original: $z = 1 + 3x_1 + 0x_2 + 1x_3 - 1x_4 + 0x_5 + 2x_6 = 8$

New: $x_1: 3 - (-3 \times 1) = 6$ $x_6: 2 - (-3 \times 2) = 11$

$x_2: 0 - (-3 \times 0) = 0$

RHS: $146 - (-3 \times 8) = 170$

$x_3: 0 - (-3 \times 1) = 3$

$x_4: 1 - (-3 \times -1) = -2$

$x_5: 0 - (-3 \times 0) = 0$

$\Rightarrow z' = 1 + 6x_1 + 0x_2 + 3x_3 - 2x_4 + 0x_5 + 11x_6 = 170$

New Optimals:

$$x_1 = 0 \quad x_2 = 14 \quad x_3 = 8$$

New Max Profit: 170

(b) (10 points) Suppose that the available amount of Resource 1 increases from 36 units to 41 units, and the available amount of Resource 3 decreases from 22 units to 20 units. How should the production plan be changed?

Original Constraints:

$$\begin{aligned} \max \quad & z = 10x_1 + 7x_2 + 6x_3 && \text{(Profit \$)} \\ \text{s.t.} \quad & 3x_1 + 2x_2 + x_3 \leq 36 && \text{(Resource 1)} \\ & x_1 + x_2 + 2x_3 \leq 32 && \text{(Resource 2)} \\ & 2x_1 + x_2 + x_3 \leq 22 && \text{(Resource 3)} \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Δ constraint: Resource 1: $3x_1 + 2x_2 + x_3 \leq 41$
 $\hookrightarrow \Delta = +5$

Resource 3: $2x_1 + x_2 + x_3 \leq 20$
 $\hookrightarrow \Delta = -2$

Original Optimal Tableau:

	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	3	0	0	1	0	5	146
x_2	0	1	1	0	1	0	-1	14
x_5	0	-2	0	0	1	1	-3	2
x_3	0	1	0	1	-1	0	2	8

→ Shadow Price Resource 1 = 1

→ Shadow Price Resource 3 = 5

Change in Obj. Func.:

$$\Delta z = (1 \times 5) + (5 \times (-2)) = -5$$

⇒ New Optimal Profit: $146 - 5 = 141$

Only x_4 and x_6 change in space, thus

$$\Rightarrow x_2 = 14 \text{ (no change)}$$

$$x_5 = 2 \text{ (no change)}$$

$$x_3 = 8 \text{ (no change)}$$

⇒ Resource 1 Optimal = $x_1 = 0$ (no change)

⇒ Resource 2 optimal = $x_2 = 14$ (no change)

⇒ Resource 3 optimal = $x_3 = 8$ (no change)

Since Resource 3 has higher shadow price our max profit ends up decreasing to 141 instead of 146 with the new change. However, our optimal production plan remains unchanged.

- (c) (10 points) Due to improvement of technology, the amount of Resource 1 needed in order to produce one unit of Product 1 is reduced from 3 units to 2 units. How should the production plan be changed?

Original Constraint Resource 1:

$$3x_1 + 2x_2 + x_3 = 36$$

New Constraint Resource 1:

$$2x_1 + 2x_2 + x_3 = 36$$

Shadow price Resource 1 = $x_4 = 1$ (unchanged)

$\Rightarrow$ Basis remains optimal, original production plan should

have no change

$$\Rightarrow x_1 = 0 \quad x_2 = 14 \quad x_3 = 8 \quad (\text{no change})$$

New Optimal Profit = 146 (no change)

$\Rightarrow$ Since Prod. 1 was not being produced in optimal solution, reducing the required Resource 1 per unit of Prod. 1 does not change optimal plan.